



Review and assessment of length-length and length-weight relationships of spiny lobsters (*Palinuridae*) captured in Brazilian waters

Lucas Vinícius Santos Silva¹ · Humber Agreli de Andrade¹

Received: 22 September 2025 / Revised: 20 November 2025 / Accepted: 24 November 2025 / Published online: 6 December 2025
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025

Abstract

Morphometric relationships are fundamental in biological and fisheries sciences, particularly for species of high commercial value such as spiny lobsters (*Panulirus* spp.). This study presents a comprehensive and systematic review of length-length (LLR) and length-weight (LWR) relationships for *P. meripurpuratus*, *P. laeviscauda*, and *P. echinatus* captured along the Brazilian coast. In addition, we also show graphically the morphometric index reported from previous studies. A total of thirteen LLR and nine LWR studies published between 1963 and 2025 were selected following PRISMA 2020 guidelines. Morphometric indices and regression equations were analyzed by sex and region. In general, males exhibited proportionally larger and heavier carapace and females present smaller and heavier abdomen, possibly reflecting sex-specific functional adaptations. Although average ratios of one-third for carapace length and two-thirds for abdomen length (AL) are frequently cited, our results show that these proportions are not stable across body sizes or sexes. Males, especially larger individuals, tend to deviate more from this pattern probably influenced by potential ecological or reproductive advantages of a large carapace. Moreover, the AL displayed greater variability as an explanatory variable, potentially influenced by morphological plasticity. Given the large number of studies reporting equations based on different body measurements, we highlighted the most recommended equations following five criteria to guide future research and applications involving LLR and LWR. These equations aim to provide a more consistent basis for comparative analyses and fisheries assessments. We list desired protocols to achieve consistency between studies, particularly for AL, and urge researchers to report regression parameters and conditions under which measurements were obtained. We also call for future studies to investigate the effects of freezing and tail positioning on morphometric accuracy.

Keywords Body proportions · Morphometric index · Sex differences

Introduction

Understanding length-length relationships (LLR) and length-weight relationships (LWR) is essential, since body size and weight are closely linked to demographic processes, such as sexual maturity, relative growth, mortality, and size at first capture (Froese and Binohlan 2000;

Froese 2006; Gubiani et al. 2020). These relationships are also pivotal for fisheries assessments and management of aquatic resources (Takács et al. 2016; Gonzalez-Martinez et al. 2020), as they allow conversions between different body measurements when some of them are not readily available. This is particularly relevant for spiny lobsters (*Panulirus* spp.), which occur worldwide in tropical and subtropical waters and occupy a wide range of habitats. These species inhabit coral reefs, rocky bottoms, and muddy substrates (Phillips et al. 2013). They also represent an important component of Brazilian commercial exports, with the tail being the main product traded (Briones-Fourzán, 2025; Gaeta et al. 2022). However, because the tail is a flexible structure, it is more susceptible to measurement variability (Alencar et al. 2020). In

✉ Lucas Vinícius Santos Silva
contatolucassantoss@gmail.com

¹ Departamento de Pesca e Aquicultura, Programa de Pós graduação em Recursos Pesqueiros e Aquicultura, Universidade Federal Rural de Pernambuco, Campus Recife, Recife, PE, Brazil

contrast, the carapace is a rigid and more morphologically stable structure, but it is not typically used in commercial trade. Therefore, body ratios are useful for estimating the total size of individuals, such as in internal and external markets.

In Brazil, three species of spiny lobsters hold substantial economic and ecological relevance in both domestic and international markets: *P. meripurpuratus* (historically referenced as *P. argus*), *P. laeviscauda*, and *P. echinatus*. Despite the widespread use of body measurement relationships for these species, the literature reveals considerable methodological heterogeneity. Studies differ in the body measurements used as explanatory variables for regression equations and in the statistical models applied (e.g., linear, exponential) to describe these relationships. Such variability in model structure, along with the lack of a unified framework connecting all response and explanatory variables across studies, can significantly affect key management decisions, especially those related to minimum legal capture sizes (Barreto et al. 2009). Currently, the minimum legal size in Brazil for *P. meripurpuratus* is 13 cm of carapace length (CL) and 7.5 cm of abdomen length (AL), whereas for *P. laeviscauda* is 11 cm CL and 6.5 cm AL (Ministério da Pesca e Aquicultura 2025). Therefore, assembling and integrating all available body measurement relationships is crucial to ensure comparability and reliability across studies.

Biometric relationships therefore represent a fundamental tool for promoting sustainable fishing practices and enhancing stock assessments for spiny lobsters. To investigate how these relationships have been applied and reported in Brazil, we conducted an integrative literature review. This method was selected because a significant portion of research on Brazilian lobster fisheries, especially older studies or those published in regional outlets, is not readily available through digital databases, limiting the applicability of bibliometric techniques. The integrative approach enables the identification of methodological gaps and trends, supporting a critical evaluation of which models and measurement practices yield the most robust and reproducible results (Snyder 2019; Booth et al. 2021). Ultimately, this contributes to refining biological parameters used in fisheries assessments and to advancing standardized protocols for future research.

Therefore, the present study aims to review and assess the biometric relationships described for spiny lobsters *P. meripurpuratus*, *P. laeviscauda*, and *P. echinatus* in Brazil. We aim to identify the most suitable model fits for application in future studies and to support more effective management practices for these fishery resources. Moreover, the results generated here may also inform the management of other spiny lobster fisheries worldwide.

Materials and methods

Review

Scientific articles were retrieved from the Google Scholar, Scopus®, and Clarivate Web of Science databases. Three keyword search strings were used to encompass the three species of spiny lobsters: (“*Panulirus argus*” OR “*Panulirus meripurpuratus*”) AND (“Brasil” OR “Brazil”); “*Panulirus laeviscauda*” AND (“Brasil” OR “Brazil”); “*Panulirus echinatus*” AND (“Brasil” OR “Brazil”). Throughout this work, we refer to *P. meripurpuratus* using both its current name and its former designation (*P. argus*) to ensure the inclusion of all historical and contemporary records. No specific keywords related to morphometry were included to ensure the retrieval of publications, regardless of language. The search was conducted within titles, abstracts, and keywords, without restrictions on publication year or language. Duplicate articles and those not addressing the size or weight of spiny lobsters captured in Brazil were excluded. Documents containing information on morphometric relationships were separated from the others and considered in the analysis. We followed the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, an internationally recognized framework for conducting systematic reviews and meta-analyses in a transparent and reproducible manner were followed for the selection of articles included in the analysis. The procedure was conducted separately for length-length relationships (LLR) and length-weight relationships (LWR). Accordingly, the article was structured into two distinct sections to facilitate a clearer understanding of each relationship type.

Measurements

Length-length relationship (LLR)

The LLR considered in this study refer to three body measurements: total length (TL), carapace length (CL), and abdomen length (AL) (Fig. 1). In most studies, TL is measured as the straight-line distance from the base of the rostral spines (the point where both spines join) to the tip of the telson, with the specimen positioned dorsally and fully extended on a flat surface. CL corresponds to the straight-line distance between the anterior margin of the supraorbital ridge (also at the base of the rostrum) and the posterior edge of the carapace along the dorsal midline. Finally, AL is measured as the straight-line distance from the anterior edge of the first abdominal somite to the posterior tip of the telson.

Although these descriptions are consistent throughout the literature reviewed here, it is important to note that TL

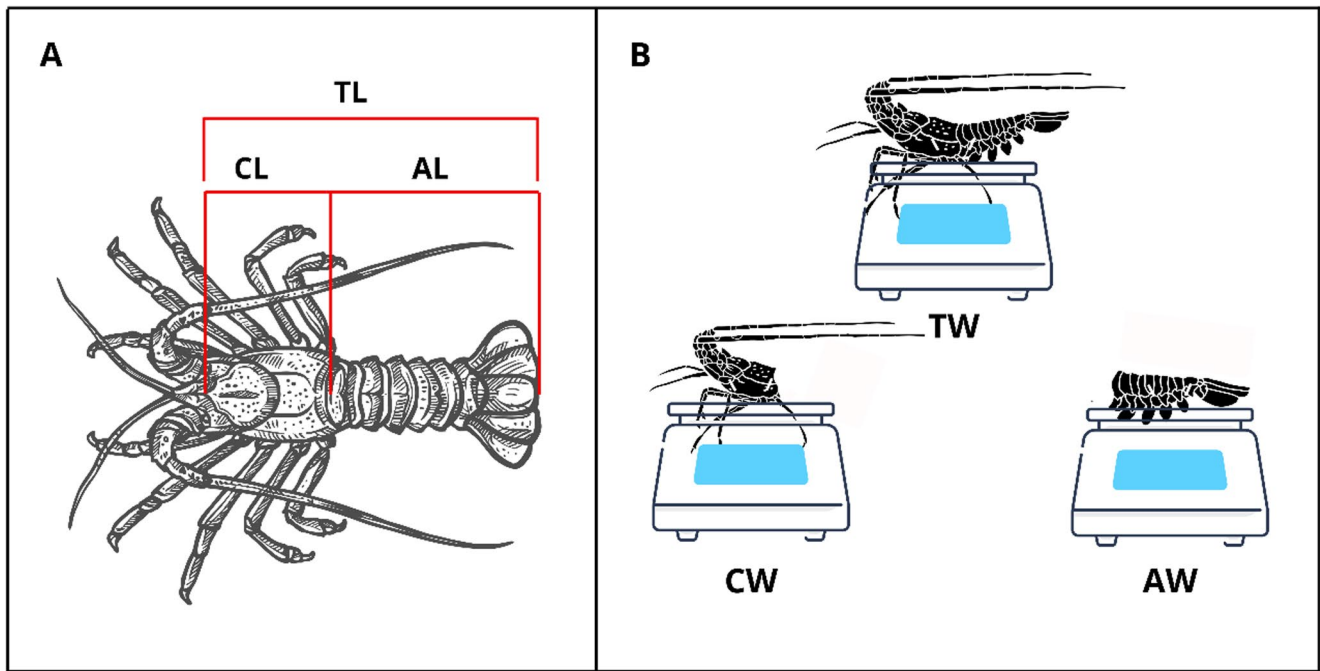


Fig. 1 Length (A) and weight (B) measurements used in the studies held for spiny lobsters captured in Brazil (TL, total length; CL, carapace length; AL, abdomen length; TW, total weight; CW, carapace weight; AW, abdomen weight)

and CL do not originate from the tip of the rostrum, as is sometimes mistakenly assumed. Instead, they start at the rostral base, which ensures greater measurement consistency. The AL measurement, however, is more complex and more prone to variability, as it depends on the precise positioning of the abdomen and the accurate identification of segmental landmarks. This complexity has been noted in previous discussions (e.g., Silva and Fonteles-Filho, 2011) and should be considered when interpreting data involving this metric. In addition, the studies varied in the type of measurements units used, reporting values in either centimeters or millimeters (Supplementary material 1 and 2).

Length-weight relationship (LWR)

The LWR that the present study focused on were the ones from three body measurements and weights (g), TL, CL, AL, total weight (TW), carapace weight (CW), and abdomen weight (AW) (Fig. 1). All studies described the weights with the following descriptions: TW is the total body weight recorded with all appendages present and intact. CW is the weight of the anterior body portion of the lobster after disarticulation at the first abdominal segment. AL is the weight of the posterior body portion of the lobster following disarticulation from the cephalothorax at the first abdominal segment. The descriptions of the body measurements were identical for every study considered here.

Morphometric Index

The morphometric index (MI) measure the percentage proportion between two measures assessed according to the expression $MI = \left(\frac{C_1}{C_2} \right) \times 100$, where C_1 represents CL or AL, and C_2 represents TL for LLW and $C_1 = CW$ or AW , and $C_2 = TW$ for LWR.

Regressions

The following analysis were separated in section length-length relationship, and length-weight relationship. A linear relationship describes a direct proportionality between two variables. In contrast, a potential relationship expresses a multiplicative association.

Length-length relationship (LLR)

Variations were found in lengths used as explanatory or response variables to estimate parameters of biometric relationships. In all studies linear regressions were used to model the relationship between the variables TL, CL and AL. All of the authors have used least squares to fit the models. Simple linear models $Y = a + bX$ were used in most of the studies, but a few of them used power function $Y = aX^b$, which Y is the response variable, X is the explanatory variable, a is the increment and b the coefficient

of allometry. In that case, least squares were applied after logarithm transformations. All models and estimations of parameters, and graphics of equations are presented in Supplementary material 1 and 3, respectively.

Length-weight relationship (LWR)

Variations were found in LWR used as explanatory or response variables to estimate parameters of biometric relationships. The majority of the studies applied a logarithmic transformation of a power model the relationship between one of the variables TL, CL, or AL, and one of the variables TW, CW, and AW. Most of the authors wrote the power model $Y = aX^b$ in the paper, but some of them (e.g. Ivo et al. 1995; Pinheiro et al. 2003; Barreto et al. 2009; Oliveira et al., 2014) wrote the linearized equation. All models and estimations of parameters, and graphics are presented in Supplementary material 2 and 4, respectively.

Results

Review

We identified 4,399 documents, including scientific articles and grey literature. After screening and excluding works that did not include spiny lobster length-length (LLR) and length-weight (LWR) relationships in Brazil, 13 studies on LLR and 9 on LWR were selected (Fig. 2).

The publication dates of these documents range from 1963 to 2020 for LLR, reflecting a peak in the 1990s for *P. meripurpuratus* and *P. laeviscauda*, and for *P. echinatus* in the 2000s. After these periods, we noticed a decline in

these studies. Among the identified studies, seven focused on both *P. meripurpuratus* and *P. laeviscauda*, two only on *P. meripurpuratus*, one only on *P. laeviscauda*, and three on *P. echinatus* (Fig. 3). For LWR, the publication dates of these documents range from 1963 to 2014, reflecting two peaks for *P. meripurpuratus* in the 1990s and 2010s, and for *P. laeviscauda* in the 1970s and 1990, and for *P. echinatus* in 2000. After the 2010s, we noticed a decline in these studies (Fig. 3). In addition, six focused on both *P. meripurpuratus* and *P. laeviscauda*, one only on *P. laeviscauda*, two only on *P. echinatus*, and one on all species.

Morphometric Indexes (MIs)

A total of three studies reported MIs for spiny lobsters captured in Brazil. All four MIs (CL/TL, AL/TL, CW/TW, and AW/TW) were calculated for *P. meripurpuratus* and *P. laeviscauda* in two studies, and for *P. echinatus* in one study. In general, males of all three species exhibited proportionally larger carapace lengths than total body length (high ratio CL/TL) compared to females, although this pattern was not statistically tested. The only exception observed was for *P. meripurpuratus* in Rios (1991). (Table 1.) Conversely, females of all three species have proportionally larger abdomens relative to total body length (AL/TL) than males (Table 1.). The only exceptions were observed for *P. meripurpuratus* in Rios (1991) and for *P. laeviscauda* in Ivo et al. (1995). Whether exceptions were observed or not, the studies do not report these differences as statistically significant. Additionally, although Ivo et al. (1995) did not report the results as statistically significant, they observed that regional variations in the analyzed indices were relatively small. Moreover, males of all three species have proportionally heavier

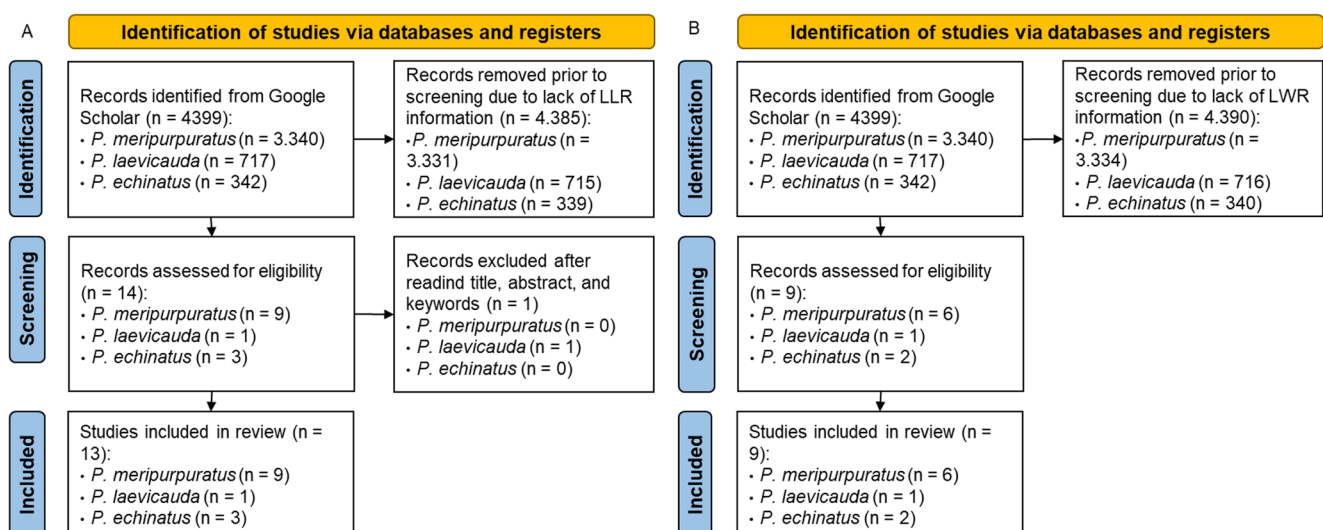


Fig. 2 PRISMA 2020 flow diagram for the identification of documents on length-length (LLR) (A) and length-weight (LWR) (B) relationships of *Panulirus meripurpuratus*, *P. laeviscauda*, and *P. echinatus* captured in Brazil

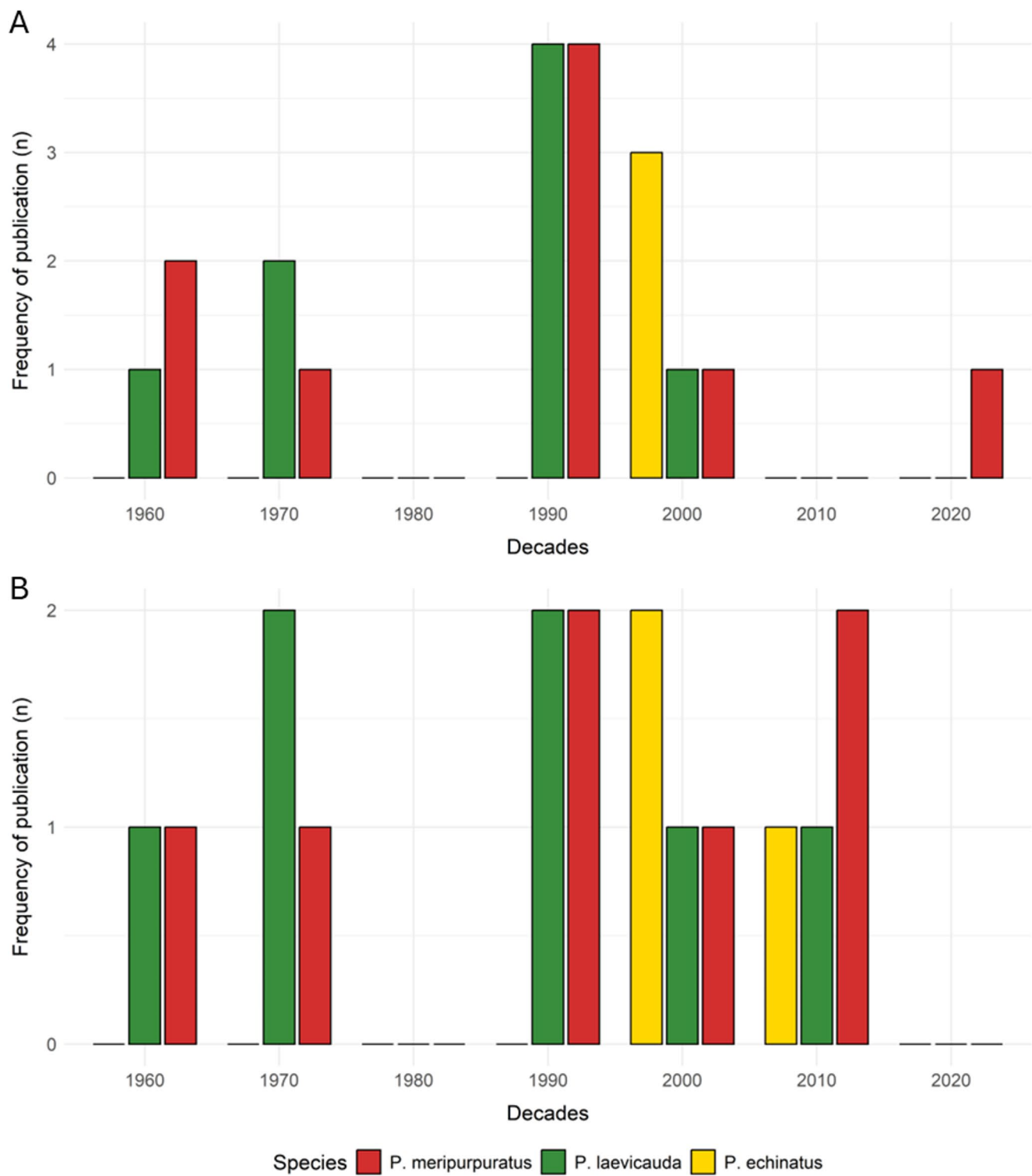


Fig. 3 Number of documents addressing length-length (**A**) and weight-length (**B**) relationships of *Panulirus meripurpuratus*, *P. laevicauda*, and *P. echinatus* captured in Brazil by decade from 1960 to 2020

Table 1 Average morphometric index (%) for *Panulirus meripurpuratus*, *P. laevicauda*, and *P. echinatus* by sex and mean (CL, cephalothorax length; TL, total length; AL, abdomen length; CW, carapace weight; TW, total weight; AW, abdomen weight; CE, Ceará state; RN, Rio Grande do Norte state; BA, Bahia state; ES, Espírito Santo state; SPSPA, Saint Peter and Saint Paul Archipelago)

Relationship	Species	Sex	Authors				
			Ivo et al. 1995			Rios 1991	Pinheiro et al. 2003
			CE	RN	BA/ES	CE	PE
CL/TL	<i>P. meripurpuratus</i>	Male	36.1	37.8	37.3	34.1	-
		Female	35.4	35.9	35.8	35.2	-
		Mean	35.7	36.8	36.5	34.6	-
	<i>P. laevicauda</i>	Male	36.5	36.9	37.7	36	-
		Female	35.5	35.8	34.8	35.5	-
		Mean	36	36.3	36.2	35.7	-
	<i>P. echinatus</i>	Male	-	-	-	-	38.7
		Female	-	-	-	-	36
		Mean	-	-	-	-	37.3
AL/TL	<i>P. meripurpuratus</i>	Male	65	65,3	66,2	57.5	-
		Female	63,8	62,3	63,3	56.5	-
		Mean	64,4	63,3	65,6	57	-
	<i>P. laevicauda</i>	Male	65	64,8	68,4	69	-
		Female	62,3	60,9	62,7	62.2	-
		Mean	63,6	62,8	65,5	65.6	-
	<i>P. echinatus</i>	Male	-	-	-	-	57.5
		Female	-	-	-	-	56.5
		Mean	-	-	-	-	57
CW/TW	<i>P. meripurpuratus</i>	Male	64.4	64,2	63,4	66.1	-
		Female	64,9	66	65.1	63.7	-
		Mean	63.4	65.1	65.3	64.9	-
	<i>P. laevicauda</i>	Male	63.8	64.7	62.1	62.9	-
		Female	65.1	64.4	64.9	63.1	-
		Mean	64.4	65.5	63.5	63	-
	<i>P. echinatus</i>	Male	-	-	-	-	62.3
		Female	-	-	-	-	64.4
		Mean	-	-	-	-	63.3
AW/TL	<i>P. meripurpuratus</i>	Male	34	33,8	33,8	29.7	-
		Female	35,3	37,3	37,3	32.9	-
		Mean	34,6	35,5	35,5	31.3	-
	<i>P. laevicauda</i>	Male	33,9	34,1	31	30.5	-
		Female	36,8	37,8	36,5	39.7	-
		Mean	35,3	35,9	33,7	35.1	-
	<i>P. echinatus</i>	Male	-	-	-	-	26.6
		Female	-	-	-	-	36.4
		Mean	-	-	-	-	31.5

abdomen than females, nonetheless, this pattern was not statistically tested. The only exception observed was for *P. meripurpuratus* in the study by Rios (1991) (Table 1). Consequently, females of all three species have proportionally heavier carapaces than males (Table 1.). The studies do not report these differences as statistically significant.

The average morphometric indices indicate an approximate proportion of one-third of TL for CL and two-thirds for AL in *P. meripurpuratus* and *P. laevicauda* (Table 1.). However, there is a trend of allometric variation, especially in males, where the carapace can represent more than one-third of the total length in larger individuals, consequently

reducing the proportional of the abdomen in the total length. The variation is less evident in females, who exhibit more stable proportions (Fig. 4a). Moreover, for males, the proportion of carapace weight remains relatively constant across the weight range in *P. laevicauda*, hovering around 70%, while in *P. meripurpuratus* it fluctuates slightly but stays around 57–60%. In contrast, the AW/TW percentage decreases with increasing total weight in both species, though *P. meripurpuratus* consistently shows slightly lower AW/TW values compared to *P. laevicauda*. Among females, both species display a declining trend in carapace weight% as total weight increases. However, *P. laevicauda*

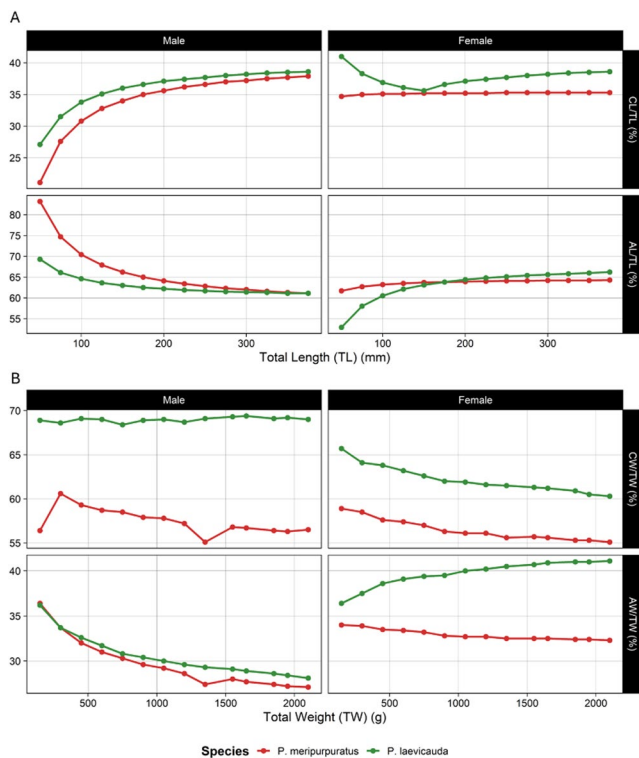


Fig. 4 Morphometric index by sex calculated by Rios (1991) for *Panulirus meripurpuratus* and *P. laevicauda* of carapace and abdomen length to total length (A) and of carapace and abdomen weight to total weight (B) (CL, carapace length; TL, total length; AL, abdomen length; CW, carapace weight; TW, total weight; AW, abdomen weight)

females start with a higher CW/TW than *P. meripurpuratus*, and both species reduce the relative investment in the carapace over time. In terms of abdomen weight, *P. laevicauda* females show a clear increasing trend. On the other hand, *P. meripurpuratus* females maintain slightly decreasing AW/TW values (Fig. 4b).

Figura 4. Morphometric index by sex calculated by Rios (1991) for *Panulirus meripurpuratus* and *P. laevicauda* of carapace and abdomen length to total length (A) and of carapace and abdomen weight to total weight (B) (CL, carapace length; TL, total length; AL, abdomen length; CW, carapace weight; TW, total weight; AW, abdomen weight).

Regressions

Among the 13 studies identified for LLR, nine used CL as the explanatory variable, eight used TL, and seven used AL. Regarding the response variables, 10 studies used TL, nine used CL and six used AL. A total of six studies was conducted in the state of Ceará, four in Pernambuco, three in Rio Grande do Norte, and one study encompassed both Bahia and Espírito Santo, while another included both Amapá and Pará (Table 2). Seven of them studied two species, *P.*

meripurpuratus and *P. laevicauda*, while *P. echinatus* was always studied separately.

For the LWR 10 studies were identified. Among these, six used CL as the explanatory variable, eight used TL, and seven used AL. Regarding the response variables, nine studies used TW, four used CW, and five used AW. Four studies were conducted in the state of Ceará, three in Pernambuco, and one in Rio Grande do Norte. Additionally, one study encompassed both Bahia and Espírito Santo, while another included both Amapá and Pará (Table 2). Six of them studied two species, *P. meripurpuratus* and *P. laevicauda*, while *P. echinatus* was always studied separately with one encompassing all species.

Length-length relationship (LLR)

Panulirus meripurpuratus

A total of nine studies were found for *P. meripurpuratus*, six of them presented TL as explanatory variable, seven with CL, and five with AL. The TL–CL and TL–AL relationships showed a strong linear association in both sexes due to high determination coefficient (Supplementary material 1), with slight regional variations in slope and in the intercept. The TL–AL relationship reported by Ivo et al. (1995), based on individuals captured in Bahia and Espírito Santo, showed a significant difference (Supplementary material 3).

Panulirus laevicauda

As observed in *P. meripurpuratus*, the TL–CL and TL–AL equations in *P. laevicauda* were consistent and highly correlated, once they had greater overlap among samples from different locations and high determination coefficient (Supplementary material 1). In general, the CL–TL relationships show high linearity and consistency across studies. The AL–TL relationship was generally linear for both sexes, but females exhibited more consistent patterns across studies (except for Vasconcelos and Vasconcelos (1994) for females), with greater overlap among regression lines. Males showed slightly more variation, particularly at the extremes (Supplementary material 3).

Panulirus echinatus

The species *P. echinatus* data revealed well-defined relationships among all length measurements. The TL–CL and TL–AL relationships showed high visual determination coefficients, indicating good predictability between measurements. The AL–TL relationship was slightly more variable (Supplementary material 3).

Table 2 Studies that analyzed length-length (LLR) and length-weight (LWR) relationships of *Panulirus meripurpuratus* (Pm), *P. laeviscauda* (Pl), and *P. echinatus* (Pe) captured in Brazil (TL, total length; CL, carapace length; AL, abdomen length; TW, total weight; CW, carapace weight; AW, abdomen weight; PE, Pernambuco; CE, Ceará; RN, Rio Grande do Norte; BA, Bahia; ES, Espírito Santo; SPSPA, Sain Peter and saint Paul Archipelago; AM, Amapá; PA, Pará; *, Jun and Dec/2001, and May, Jul, Aug, Oct, and Nov/2002)

Relationship	Authors	Species	Explanatory length variable	Explanatory weight variable	Sampling area	Sampling period
LLR	Coelho and Moura 1963	Pm; Pl	TL; CL	TL; CL	PE	Jun/1961 - Dec/1961
	Santos et al. 1964	Pm	TL		PE; CE	Jan/1962 - Dec/1963
	Rolim and Rocha 1972	Pm; Pl	TL; AL	TL; AL	CE	Oct/1964 - Oct/1972
	Nascimento 1976	Pl	TL	TL	CE	Mar/1972 - Jan/1973
	Rios 1991	Pm; Pl	TL; CL; AL	TL; CL; AL	CE (Fortaleza)	Sep/1989 - Aug/1991
					CE (Almofala)	Nov/1989 - Apr/1991
	Castro 1992	Pm; Pl	CL		CE	Mar/1990 - Apr/1992
	Vasconcelos and Vasconcelos 1994	Pm; Pl	CL; AL		RN	May/1993 - Dec/1993
	Ivo et al. 1995	Pm; Pl	TL; CL; AL	TL; CL; AL	CE; RN; BA and ES	May/1994 - Jun/1995
	Rocha and Xavier 2000	Pm; Pl	CL; AL	CL; AL	NE	Jan/1965 - Dec/1972
	Silva et al. 2001	Pe	CL; AL		RN (Rocas Atoll)	Jul/2000 - Mar/2001
	Pinheiro et al. 2003	Pe	TL; AL	TL; AL	PE (SPSPA)	Jun/1998 - Sep/1999
	Barreto et al. 2009	Pe	CL	CW	PE	Nov/1999 - Oct/2000
	Oliveira et al., 2014	Pm; Pl; Pe		TW	PB	Apr/2006 - Sep/2007
	Peixoto et al. 2020	Pm	TL; CL	TL; CL	AM; PA	Jun and Dec/2001, and May, Jul, Aug, Oct, and Nov/2002
LWR	Coelho and Moura 1963	Pm; Pl	TL; CL	TW	PE	Jun/1961 - Dec/1961
	Rolim and Rocha 1972	Pm; Pl	TL; CL; AL	TW; CW; AW	CE	Oct/1964 - Oct/1972
	Nascimento 1976	Pl	TL	TW	CE	Mar/1972 - Jan/1973
	Rios 1991	Pm; Pl	TL; CL; AL	TW; CW; AW	CE (Fortaleza)	Sep/1989 - Aug/1991
					CE (Almofala)	Nov/1989 - Apr/1991
	Ivo et al. 1995	Pm; Pl	TL; CL; AL	TW; CW; AW	CE; RN; BA and ES	May/1994 - Jun/1995
	Rocha and Xavier 2000	Pm; Pl	TL; CL	TW; AW	NE	Jan/1965 - Dec/1972
	Pinheiro et al. 2003	Pe	TL	TW	PE (SPSPA)	Jun/1998 - Sep/1999
	Barreto et al. 2009	Pe	TL; CL; AL	TW; CW; AW	PE	Nov/1999 - Oct/2000
	Oliveira et al., 2014	Pm; Pl; Pe	TL	TW	PB	Apr/2006 - Sep/2007

Length-weight relationship (LWR)

Panulirus meripurpuratus

The TL–TW relationship showed a strong linear association in both sexes as indicated by high determination coefficient, with slight regional variations in slope higher in females than in males (Supplementary material 2 and 4). Males exhibit higher CW than females for the same TL, particularly at intermediate and larger sizes. In the CL–CW relationship, more pronounced differences are observed across authors. The curves from Ivo et al. (1995) show notably higher cephalothorax weights at larger sizes compared to earlier studies like Rolim e Rocha (1972). Between sexes, males consistently exhibit higher cephalothorax weights than females, especially at larger carapace lengths. The AL–AW relationship presents the most consistent and comparable patterns. All studies from Ivo et al. (1995) display nearly overlapping curves for both males and females, indicating minimal sexual dimorphism in abdominal weight once adjusted

for abdomen length. However, Ivo et al. (1995) (3) shows slightly higher values, especially for females.

Panulirus laeviscauda

For TL–TW, male specimens show marked variation, with the curves from Ivo et al. (1995) (3), indicating the highest TW for a given TL. Among females, similar trends are observed, with Ivo et al. again yielding higher weight estimates, especially beyond 200mm TL. In the CL–CW panel, data across authors are generally consistent, with Rios (1991) and Rolim and Rocha (1972) presenting nearly identical curves for both sexes. For AW, females show slightly steeper curves than males, particularly in the datasets from Rocha and Xavier (2000) and Oliveira et al. (2014).

Panulirus echinatus

For both sexes, the curves from Pinheiro et al. (2003) are consistently positioned above those of Oliveira et al. (2014),

indicating higher predicted weights for a given total length. Among females, the difference between the two datasets is particularly evident at larger sizes (>150 mm TL), where the weight estimates from Pinheiro et al. (2003) diverge substantially from those of Oliveira et al. (2014), reaching values above 500 g. In males, the same trend is observed, with Pinheiro et al. (2003) again predicting significantly higher weights for equivalent lengths. The divergence becomes more pronounced as TL increases. For TL–TW, males attain higher weights than females, particularly at larger carapace lengths. A similar pattern is observed for CL–CW, where male curves consistently exceed those of females across the entire length range. In contrast, the pattern reverses for AL–AW. Females display markedly higher AW values than males throughout the observed size range.

Given the large number of studies reporting equations based on different body measurements, we selected and highlighted the most suitable equations to guide future research and applications involving length–length and length–weight relationships. The selection of recommended equations followed a hierarchical procedure that prioritized: (i) coverage of length ranges, (ii) spatial representativeness of sampling areas, (iii) sample size, and (iv) temporal coverage. Equations based on broader and more representative datasets were favored to maximize applicability across ontogenetic stages and fishing grounds, while larger and temporally extended samples were prioritized when other criteria were comparable. Determination coefficients (r^2) were considered only as complementary information, since they were very similar among studies. The recommended equations, presented in Table 3, provide a more robust and standardized basis for comparative analyses and fisheries assessments.

Discussion

Morphometric Index (MI)

Regarding the MI, differences between sexes indicate that females generally possess proportionally larger and heavier abdomens. Males, in contrast, tend to show larger and heavier carapace than females, particularly in larger individuals (Rios 1991). These findings corroborate previous studies for lobsters in Brazil, although statistical testing of these differences is often lacking, and other crustaceans (Ivo et al. 1995; Vogt 2016). Data from Rios (1991) revealed dynamic trends: among males, CW% remained relatively constant in *P. laeviscauda* (~70%) but fluctuated around 57–60% in *P. meripurpuratus*. Among females, CW% declined with size in both species, while AW% increased in *P. laeviscauda*, but decreased slightly in *P. meripurpuratus*.

Regarding the males: as total length increases, the carapace tends to represent a larger proportion of body length, often exceeding one-third, while the relative length of the abdomen decreases. These trends, however, were not statistically tested in the reviewed studies and should be interpreted as descriptive observations rather than statistically significant differences.

This raises the question of whether males with proportionally larger carapaces might gain functional advantages beyond mere morphological differentiation. A larger carapace may offer enhanced protection during agonistic interactions or act as a visual signal in sexual selection, potentially increasing male success in territorial disputes or mate acquisition. Unlike females, which invest in abdominal development to facilitate egg incubation, males are free from such reproductive constraints. This may enable greater energy allocation toward carapace growth or secondary sexual traits such as enlarged chelae. This is also observed for other crustaceans (e.g., Fadlaoui et al. 2019; Pacheco et al. 2021). However, these advantages must be balanced against the metabolic and locomotor costs of increased cephalothorax-to-body ratio. Further behavioral and ecological studies are needed to determine whether these morphometric patterns translate into actual fitness advantages in natural populations.

Regressions

Length-length relationship (LLR)

The frequent citation of Rocha and Xavier (2000) by subsequent authors (Soares and Cavalcante 1985; Oliveira and Pereira 1982; Salazar and Cruz 2019), despite the unavailability of the original manuscript and the absence of essential methodological details (e.g., length range, coefficient of determination) in the published version, raises concerns about reproducibility. This highlights the need for more rigorous methodological reporting in future studies, including specimen size ranges, sample sizes, regression parameters estimates, and goodness-of-fit metrics.

Most studies used simple linear regressions, while others applied power functions or log-transformed power models (Rocha and Xavier 2000; Pinheiro et al. 2003). Despite these methodological differences, strong correlations were consistently observed among TL, CL, and AL, suggesting stable allometric scaling across measurements. This consistency suggests that general trends in body proportions are maintained across geographic regions and sampling efforts. The greatest AL–TL and AL–CL variabilities may be partially attributed to the positioning of the tail during data collection, since abdominal length is more susceptible to flexion or extension. Additional sources of variability

Table 3 Regressions of length-length relationships of *Panulirus meripurpuratus* (Pm), *P. laevicauda* (Pl), and *P. echinatus* (Pe) captured in Brazil (TL, total length; CL, carapace length; AL, abdomen length; TW, total weight; CW, carapace weight; AW, abdomen weight)

Regression	Species	Male	Author	Female	Author
TL - CL	Pm	CL = −10.63490+0.41804 TL	Ivo et al. 1995 (2)	CL = −0.95814+0.35904 TL	Ivo et al. 1995 (2)
	Pl	CL = −3.16404+0.38980 TL	Ivo et al. 1995 (2)	CL = 5.46+0.3261 TL	Rios 1991 (2)
	Pe	CL = 0.243TL ^{1.16}	Pinheiro et al. 2003	CL = 0.296TL ^{1.07}	Pinheiro et al. 2003
CL - TL	Pm	lnTL = 1.61+0.86 lnCL	Rocha and Xavier 2000	TL = 7.63698+2.73252 CL	Ivo et al. 1995 (2)
	Pl	TL = 27.69744+2.33181 CL	Ivo et al. 1995 (3)	lnTL = 1.33+0.93 lnCL	Rocha and Xavier 2000
	Pe	TL = 0.182CL ^{1.148}	Barreto et al. 2009	TL = 0.307CL ^{1.030}	Barreto et al. 2009
TL - AL	Pm	AL = 23.21201+0.54672 TL	Ivo et al. 1995 (3)	AL = −5.17049+0.67963 TL	Ivo et al. 1995 (2)
	Pl	AL = 2.29+0.6130 TL	Rios 1991 (2)	AL = −1.09192+0.67076 TL	Ivo et al. 1995 (2)
	Pe	AL = 0.838TL ^{0.889}	Pinheiro et al. 2003	AL = 0.703TL ^{0.968}	Pinheiro et al. 2003
AL - TL	Pm	lnTL = 0.39+1.01 lnAL	Rocha and Xavier 2000	lnTL = 0.71+0.94 lnAL	Rocha and Xavier 2000
	Pl	TL = −10.53030+1.66875 AL	Ivo et al. 1995 (1)	TL = −5.43385+1.58839 AL	Ivo et al. 1995 (1)
	Pe	TL = −0.8175+1.7482 AL	Silva et al. 2001	TL = −0.0141+1.6097 AL	Silva et al. 2001
TL - TW	Pm	TW = 0.000022 TL ^{3.12424}	Ivo et al. 1995 (2)	TW = 0.000052 TL ^{2.95027}	Ivo et al. 1995 (2)
	Pl	TW = 0.000084 TL ^{2.87075}	Ivo et al. 1995 (2)	TW = 0.000135 TL ^{2.76606}	Ivo et al. 1995 (2)
	Pe	TW = 0.00002 TL ^{3.1779}	Pinheiro et al. 2003	TW = 0.00004 TL ^{2.9649}	Pinheiro et al. 2003
CL - CW	Pm	CW = 0.000735 CL ^{2.92246}	Ivo et al. 1995 (2)	CW = 0.000833 CL ^{2.93037}	Ivo et al. 1995 (2)
	Pl	CW = 0.001364 CL ^{2.7785}	Ivo et al. 1995 (2)	CW = 0.000986 CL ^{2.85276}	Ivo et al. 1995 (2)
	Pe	CW = 0.001CL ^{2.978}	Barreto et al. 2009	CW = 0.001CL ^{2.849}	Barreto et al. 2009
AL - AW	Pm	AW = 0.000171 AL ^{3.78058}	Ivo et al. 1995 (2)	AW = 0.000191 AL ^{2.74491}	Ivo et al. 1995 (2)
	Pl	AW = 0.000076 AL ^{2.93566}	Ivo et al. 1995 (2)	AW = 0.0002571 AL ^{2.66454}	Ivo et al. 1995 (2)
	Pe	AW = 0.0001AL ^{3.147}	Barreto et al. 2009	AW = 0.0001AL ^{2.913}	Barreto et al. 2009

include preservation procedures (e.g., freezing) and biological factors such as growth phase or sex.

Length-weight relationship (LWR)

Length-weight relationships are fundamental in fisheries science because they provide insights into growth patterns, condition, and population structure, and allow biomass estimation from length data for management and stock assessment purposes (Froese 2006). The TL–TW curves showed substantial variation among studies. For instance, Ivo et al. (1995) (particularly dataset 3) and

Rios (1991) (dataset 2) predicted higher TW at larger TLs in comparison to Rolim and Rocha (1972), Rocha and Xavier (2000) and Oliveira et al. (2014). CL–CW curves were often similar between sexes, suggesting weak sexual dimorphism in cephalothorax growth. The analysis of AL–AW highlighted the clearest evidence of dimorphism. Ivo et al. (1995) (especially dataset 3) and Barreto et al. (2009) showed that females accumulated substantially more abdominal weight than males at equivalent sizes, reflecting their reproductive role (Supplementary material 4) These differences were consistent across the entire observed length range and should be carefully considered

when analysing reproductive variables (e.g. fecundity) or conducting energetic modeling.

Recommendations

The spiny lobster export chain frequently involves the freezing of specimens (Alencar et al. 2020). However, the extent to which freezing affects the accuracy of body measurements in lobsters remains uncertain, whether due to tissue shrinkage, increased rigidity of the exoskeleton after freezing, or the absence of a standardized protocols for the degree of abdominal extension or compression against a flat surface during measurements. Consequently, morphometric conversions derived from frozen individuals may differ from those obtained from fresh specimens, potentially introducing bias in size-based regulation enforcement and stock assessments. To support practical applications, we compiled all representative regression equations from literature into a supplementary file, providing reference models for studies requiring morphometric conversion. We recommend consulting Table 3 to select equations appropriate to each species, sex, and measurement conditions.

In light of these findings, we propose the following recommendations for future studies: (a) standardize measurement protocols across studies, particularly for AL, to reduce variability associated with preservation methods and tail positioning during data collection; (b) report complete details of regression model, including coefficients of determination, parameter estimates, confidence intervals, sample sizes, and whether measurements were obtained from fresh or frozen specimens; (c) statistically evaluate morphometric differences between sexes, size classes, and preservation conditions using appropriate tests (e.g., ANCOVA, Student's test, or non-parametric alternatives); and (d) develop correction factors to account for the effects of freezing and storage on lobster body proportions.

Overall, this synthesis reinforces the need to account for sexual dimorphism, methodological differences, and study-specific contexts when applying LLR and LWR in ecological analyses, stock assessments, and biomass estimations. The morphometric information and regression models compiled here provide a practical foundation for fisheries management, enabling the conversion of length data into biomass estimates, supporting assessments of stock composition, monitoring compliance with minimum size regulations, and improving yield predictions. In addition, morphometric indices can reveal sex-specific growth trajectories and environmentally driven variations, offering valuable insights into population structure and condition. Together, these findings underscore the relevance of integrating morphometric relationships into management frameworks to enhance the accuracy, flexibility, and biological realism of decision-making in spiny lobster fisheries.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s41208-025-01011-1>.

Author Contributions L.V.S.S wrote the main manuscript text and analyzed the data and H.A.A reviewed the manuscript and the script.

Funding The first author received financial support from the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), under grant number 88887.912927/2023-00.

Data Availability Data is provided within the manuscript or supplementary information files.

Declarations

Competing Interests The authors declare no competing interests.

References

- Alencar CAG, Tavares LS, Cintra IHA (2020) Current state of Lobster exports in Brazil. *Research, society and Development*, v. 9. 8. <https://doi.org/10.33448/rsd-v9i8.5804>
- Barreto AV, Lourdes Zani-Teixeira M, Ivo CTC, Katsuragawa M (2009) Biometric relationships of the spotted lobster, *Panulirus echinatus*, from Tamandaré coastal reefs, Pernambuco State, Brazil. *J Mar Biol Assoc United Kingd* 89(8):1601–1606
- Booth A, Martyn-St James M, Clowes M, Sutton A (2021) Systematic approaches to a successful literature review
- Briones-Fourzán P (2025) Influence of habitat enhancement and habitat degradation on spiny lobster populations in the Caribbean: an overview. *Fish Res* 281:107193. <https://doi.org/10.1016/j.fishres.2024.107193>
- Silva, A. C. D., & Fonteles-Filho, A. A. (2011). Avaliação do defeso aplicado à pesca da lagosta no nordeste do Brasil. Editora Expressão Gráfica, Fortaleza.
- Oliveira, P. A., Crispim, M. C. B., & Lúcia, A. (2014). Caracterização populacional das lagostas do gênero *Panulirus* nos ambientes recifais da Praia do Seixas e da Penha-PB. *Gaia Scientia*, 8(1), 365–383.
- Castro NMD (1992) Biometria e desenvolvimento dos caracteres sexuais secundários em lagostas jovens do gênero *Panulirus* white
- Coelho PA, Moura SJC (1963) Nota sobre as relações existentes entre as Medidas de comprimento e peso Das Lagostas *Panulirus Argus* (Latreille) (Decapoda: Palinuridae) no litoral do Estado de Pernambuco. *Trab Inst Oceanogr Univ Fed Pe* 3(1):53–60
- Fadlaoui S, Mahjoub M, El Asri O, Melhaoui M (2019) Allometric growth of the freshwater crab *Potamon algeriense* (Bott, 1967) (Decapoda, Brachyura, Potamidae) in Oued Zegzel, a mountain stream, in the northeast of Morocco. *Int J Zool* 2019:e5168639. <https://doi.org/10.1155/2019/5168639>
- Froese R (2006) Cube law, condition factor and weight-length relationships: history, meta-analysis and recommendations. *J Appl Ichthyol* 22(4):241–253. <https://doi.org/10.1111/j.1439-0426.2006.00805.x>
- Froese R, Binohlan C (2000) Empirical relationships to estimate asymptotic length, length at first maturity and length at maximum yield per recruit in fishes, with a simple method to evaluate length frequency data. *J Fish Biol* 56(4):758–773
- Gaeta J, Cintra IA, Torres MT, Silva D, Cruz R (2022) Spiny lobster fecundity on the Brazilian continental shelf (Decapoda, Achelata). *Crustaceana* 95(5–6):593–604. <https://doi.org/10.1163/15685403-bja10208>

- Gonzalez-Martinez A, Lopez M, Molero HM, Rodriguez J, González M, Barba C, García A (2020) Morphometric and meristic characterization of native Chame fish (*Dormitator latifrons*) in Ecuador using multivariate analysis. *Animals* 10(10):1805. <https://doi.org/10.3390/ani10101805>
- Gubiani EA, Ruaro R, Ribeiro VR, Santa Fé UMG (2020) Relative condition factor: Le cren's legacy for fisheries science. *Acta Limnol Bras* 32(3):e3. <https://doi.org/10.1590/S2179-975X13017>
- Ivo CTC, Vasconcelos EMS, Magalhaes JAD (1995) Avaliação dos parâmetros biométricos das lagostas *Panulirus argus* (Latreille) e *Panulirus laeviscauda* (Latreille), na plataforma continental nordeste/sudeste do Brasil. *Bol Téc Cient CEPENE* 3:169–200
- Ministério da Pesca e Aquicultura (2025) *Safra da Lagosta 2025: limite de captura, monitoramento e controle*. <https://www.gov.br/mpa/pt-br/assuntos/noticias/safra-da-lagosta-2025-limite-de-captura-monitoramento-e-controle>
- Nascimento IV (1976) Sobre a reprodução da lagosta *Panulirus laeviscauda* (Latr.). *SUDENE - Série Estudos de Pesca* nº 1
- Phillips BF, Melville-Smith R, Kay MC, Vega-Velázquez A (2013) *Panulirus* species. Lobsters: biology, management, aquaculture and fisheries, 289–325. <https://doi.org/10.1002/9781118517444.ch10>
- Pinheiro AP, Freire FAM, Lins-Oliveira JE (2003) Population biology of *Panulirus echinatus* Smith, 1869 (Decapoda: Palinuridae) from São Pedro e São Paulo Archipelago, Northeastern Brazil. *Nauplius* 11(1):27–35
- Rios GJ (1991) Análise dos caracteres biométricos das lagostas *Panulirus argus* (Latreille) e *Panulirus laeviscauda* (Latreille), no estado do Ceará, Brasil
- Rocha CAS, Xavier AFS (2000) Relações biométricas das lagostas espinhosas *Panulirus argus* (Latreille) e *Panulirus laeviscauda* (Latreille) do Nordeste do Brasil
- Rolim AE, Rocha CAS (1972) Biometria de lagostas jovens do gênero *Panulirus* Gray. *Arq Ciên Mar* 12(2):91–72
- Oliveira GM, Pereira HL (1982) La pesqueria de langosta en el nordeste de Brasil. *WECAF Reports*, n. 36, pp. 79–113
- Pacheco C, Cusba J, Paramo J, Queirolo D, Pérez D (2021) Spatial structure and morphometric relationships of the deep-sea shrimp *Solenocera acuminata* (Decapoda, Solenoceridae) in the Colombian Caribbean. *ZooKeys* 1040:1–24. <https://doi.org/10.3897/zookeys.1040.61005>
- Peixoto UI, Mesquita EM, Cintra IA, Klautau AG, Gouveia NA, Paes ET, Isaac VJ (2020) Population dynamics and sustainability of the spiny lobster (*Panulirus meripurpuratus* Giraldez & Smyth, 2016) fishery on the Amazon continental shelf. *Mar Freshw Res* 72(1):99–109
- Salazar ISO, Cruz R (2019) Age and growth of the commercial spiny lobsters *Panulirus argus* (Latreille, 1804) and *Panulirus laeviscauda* (Latreille, 1817) (Decapoda, Palinuridae) in Northeast Brazil: a review. *Crustaceana* 92(7):767–788
- Santos EPD, Costa RSD, Moura JCD (1964) Growth of the spiny Lobster *Panulirus Argus* (Latr.): quantitative aspect. *Arq Ciên Mar* 4(2):41–44
- Silva MB, Campos CEC, Targino SG, da Costa Arcoverde CED (2001) Aspectos populacionais da lagosta pintada *Panulirus echinatus* Smith, 1869 na Reserva Biológica do Atol das Rocas–Brasil. *Holos Environment* 1(2):187–198
- Soares CNC, Cavalcante PL (1985) Caribbean spiny Lobster (*Panulirus argus*) and smoothtail spiny Lobster (*Panulirus laeviscauda*) reproductive dynamics on the Brazilian Northeastern Coast. *FAO Fish Rep N* 327:200–217
- Snyder H (2019) Literature review as a research methodology: an overview and guidelines. *J Bus Res* 104:333–339
- Takács P, Vítál Z, Ferincz Á, Staszny Á (2016) Repeatability, reproducibility, separative power and subjectivity of different fish morphometric analysis methods. *PLoS One* 11(6):e0157890. <https://doi.org/10.1371/journal.pone.0157890>
- Vasconcelos JA, Vasconcelos EMS (1994) Determinação de novos parâmetros biométricos das lagostas *Panulirus argus* (Latreille) e *Panulirus laeviscauda* capturadas no litoral do Rio Grande do Norte. *Boletim Técnico Científico do CEPENE* 2(1):51–58
- Vogt G (2016) Structural specialties, curiosities, and record-breaking features of crustacean reproduction. *J Morphol* 277(11):1399–1422

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.